Solar Energy Materials & Solar Cells 160 (2017) 275–286

Contents lists available at ScienceDirect

Solar Energy Materials & Solar Cells

journal homepage: www.elsevier.com/locate/solmat

Cooling methodologies of photovoltaic module for enhancing electrical efficiency: A review

A. Shukla[a], Karunesh Kant[a,*], Atul Sharma[a], Pascal Henry Biwole[b,c]

[a] Non-Conventional Energy Laboratory, Rajiv Gandhi Institute of Petroleum Technology, Raebareli, UP India
[b] University of the Côte d'Azur, CNRS, J.A. Dieudonné laboratory, France
[c] Mines Paris Tech, PSL Research University, Center for Processes, Renewable Energies and Energy Systems, Sophia Antipolis, France

ARTICLE INFO

Keywords:
Temperature regulation
Phase change material
Photovoltaic (PV) cell
Latent heat
Heat pipe

ABSTRACT

Solar Photovoltaic (PV) cells can absorb up to 80% of the incident solar radiation obtained from the solar band, however, only a small amount of this absorbed incident energy is transformed into electricity depending on the conversion efficiency of the PV cells and part of remainder energy increases the temperature of PV cell. High solar radiation and ambient temperature lead to an elevated photovoltaic cell operating temperature, which affects its lifespan and power output adversely. Number of techniques have been attempted to maintain the temperature of photovoltaic cells close to their nominal operating value. In the present review various cooling techniques such as natural and forced air cooling, hydraulic cooling, heat pipe cooling, cooling with phase change materials and thermoelectric cooling of PV panels are discussed at length. It is important to note that, though cooling techniques are highly needed to regulate the PV module temperature, especially for mega installations, these should be economically viable too.

1. Introduction

The growing global demand for electricity, rising fossil fuel prices and increasing apprehensions about global warming have reenergized the idea to rapidly move towards renewable energy resources, especially during last two decades. Solar energy is the most abundant energy source available on the earth. Though technologies for converting sunlight energy to power have made a lot of progress, high capital price and low conversion proficiency are the main obstacles to the common use of these technologies. In order to increase the efficiency of solar power generation and make it a more cost-effective technology, different approaches have been attempted over the years consistently.

Photovoltaic (PV) is a method of transforming sunlight into electricity with the help of semiconductor materials that show the photovoltaic effect. Photovoltaic systems are composed of a number of solar cells to supply serviceable solar power. Power generation from solar PV is a clean and sustainable tool, which depends upon the planet's most ample and widely spread renewable energy source of the sun. Straight transformation of sunlight to electricity takes place without any moving parts or environmental releases through the process. This technology is well recognised, as photovoltaic systems have now been used for over fifty years both for stand-alone uses, and grid-connected installations. It is well known that the efficiency of photovoltaic solar cells declines with an increase in temperature. This decline is determined first of all by the drop of open- circuit cell voltage, and therefore, an efficient performance of PV cells in conditions, for example, of concentrated sunlight, needs to be cooled. Earlier theoretical studies [1–5] have shown the efficiency decrease which is expected, but the carrier transport mechanism put a significant effect on the actual temperature coefficient of the efficiency, and in different cases could differ significantly depending on the ambient conditions; the effects of recombination generally increase the efficiency deviation with temperature.

2. Temperature effect on photovoltaic cells

PV cells absorb 80% of incident solar radiation, but they do not fully convert it into electricity. The conversion efficiency depends upon the PV cell technology used. The remainder part of solar radiation increase solar cell temperature up to 40 °C above the atmospheric temperature [6]. This is due to the fact that PV cells convert a range of definite wavelength of the solar spectrum of light into electricity and the rest of the incident solar spectrum is converted into heat [7]. The conversion efficiency of single junction solar cells ranges between 6% and 25% under optimal operating conditions, which depends upon the semiconductor material used for the preparation of the solar cell [8,9].

* Corresponding author.
E-mail addresses: amritanshu.shukla@gmail.com (A. Shukla), k1091kant@gmail.com (K. Kant).

http://dx.doi.org/10.1016/j.solmat.2016.10.047
Received 28 July 2016; Received in revised form 7 October 2016; Accepted 26 October 2016
Available online 04 November 2016
0927-0248/ © 2016 Elsevier B.V. All rights reserved.

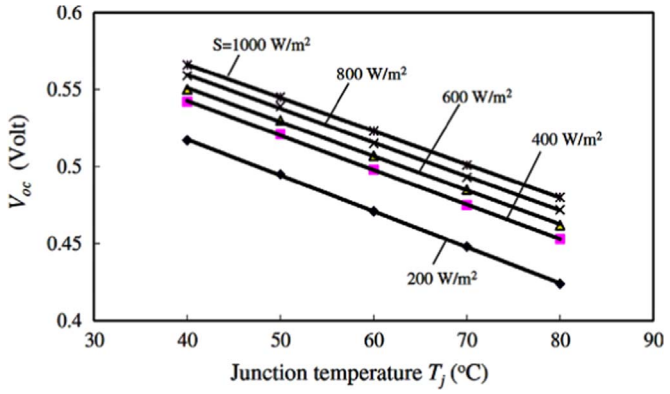


Fig. 1. Variation of open-circuit voltage with a junction temperature of PV cell [13].

Various elements affect the performance of PV modules which are installed in outdoor locations. Influences of low irradiance, dust accumulation on PV panels, and elevated operating temperatures leads to the reduction of photovoltaic conversion efficiency and technical lifetime [10,11]. The efficiency of PV cell decreases due to increase in intrinsic carrier concentrations at higher temperatures, which tend to upsurge the dark saturation current of the p–n junction [7,12]. Decrease in band-gap due to high doping also helps to upsurge the intrinsic carrier concentration [7]. The increase in dark saturation current causes the open-circuit voltage to decrease linearly which for silicon at 300 K corresponds to about −2.3 mV/°C [7]. Huang et al. [13] performed an experimental investigation to see the effect of operating temperature on the variation of open-circuit voltage with temperature (40–80 °C) at different solar radiation (200–1000 W/m^2), Fig. 1.

Fig. 2(a, b) shows the variation of power output with voltage for different temperatures. The short circuit current, increases up to some extent due to drop in the band-gap energy, Fig. 3. Andreev et al. [14] estimated an increase in the short circuit current of 0.1%/°C due to a reduction in the bandgap of the solar cell with temperature varying between 20 and 100 °C. In spite of this increase in current, the degradation of the open-circuit voltage leads to an evident decrease in the available maximum electrical power which can be better observed in the characteristic curves of PV modules at different operating temperatures in Fig. 3. For crystalline silicon PV cells, a drop in the electrical power output of about 0.2–0.5% is seen with an increase of 1 °C PV module temperature, principally due to the temperature dependence of the open-circuit voltage of the cell [15,16]. This property of PV cells is known as the temperature coefficient. According to Del Cueto [17], the decrease in efficiency due to temperature is in the range of e 1–2% for PV panel lying above a temperature span of 30 °C. Table 1 presents the temperature coefficients of various PV technologies along with their typical efficiencies [18–21].

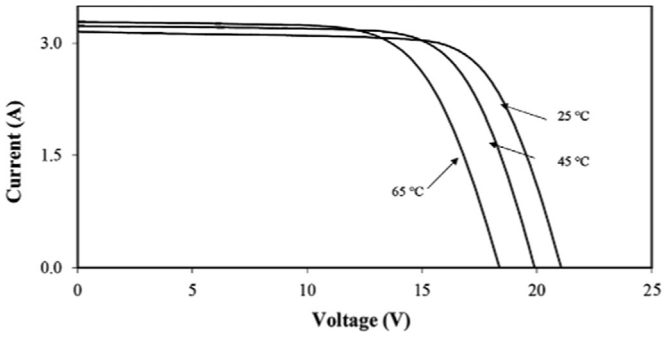


Fig. 3. Influence of temperature on PV module I–V curve [14].

Table 1
Temperature coefficients of different PV cell technologies.

T_{ref} (°C)	ηT_{ref} (%)	βref (°C^{-1})	PV technology	Refs.
25	16–24	0.0041	Mono-cSi	[20]
25	14–18	0.004	Poly-cSi	[21]
25	4–10	0.011	a-Si	[21]
25	7–12	0.0048	CIS	[17]
25	10–11	0.00035	CdTe	[17]

3. Temperature regulation systems: a review

The temperature regulation of photovoltaic panels can be achieved by several methods such as air cooling, water cooling, use of heat pipe, phase change materials and thermoelectric cooling. Some of these methodologies have been excellently reviewed by chow et al. [22] and Moradi et al. [23]. In the present study, all these available cooling methodologies (have been reviewed and the merits and drawbacks of these cooling techniques with a comparable outlook are presented Summary of the same is also given in tabular form in Table 2.

3.1. Air cooling

3.1.1. Natural air circulation

Air-cooling with a natural flow due to stack effect represents a non-expensive and simple method of PV cooling. In the natural air cooling, no extra arrangement is required for cooling PV panels. However, systems with heat extraction by natural air circulation are limited in their thermal performance due to the low density, the lower volumetric heat capacity and thermal conductivity of air and measures for heat

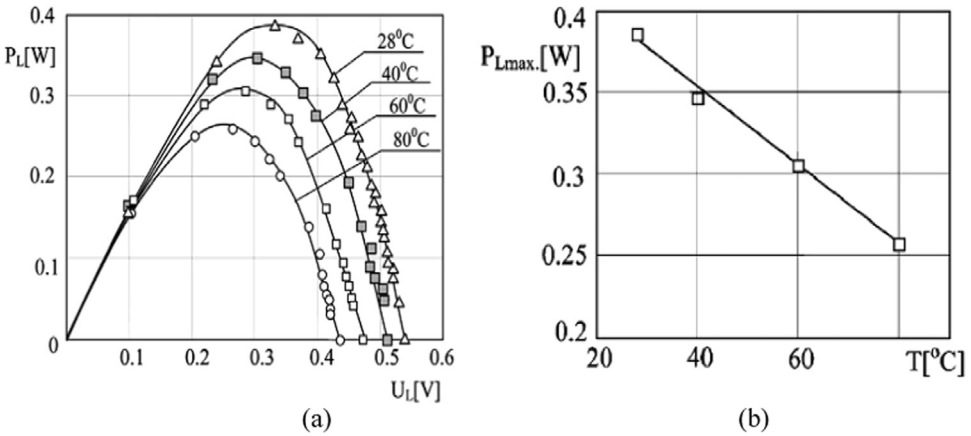


Fig. 2. The output power of single-crystalline silicon PV cells at different operating temperatures (a) temperature dependence of the maximum output power (b) [9].

Table 2
Merits and drawbacks of different thermal management techniques [45].

	Natural air circulation	Forced air circulation	Hydraulic cooling	Heat pipes	PCM thermal management	Thermoelectric (Peltier) cooling
Merits	Low initial cost, no maintenance, easy to integrate, longer life, no noise, no electricity consumption, passive heat exchange	Higher heat transfer rates compared to natural circulation of air, independent of wind direction and speed, higher mass flow rates than natural air circulation, higher temperature reduction compared to natural air circulation	Higher heat transfer rate compared to natural and forced circulation of air, higher mass flow rates compared to natural and forced circulation of air, higher thermal conductivity and heat capacity of water compared to air, higher temperature reduction	Passive heat exchange low cost, easy to integrate.	Higher heat transfer rates compared to both forced air circulation and forced water circulation, higher heat absorption due to latent heating, isothermal heat removal, no electricity consumption, passive heat exchange, no noise, no maintenance cost, heat delivery on demand	no moving parts, noise free, small size, easy to integrate, low maintenance costs, solid state heat transfer
Drawbacks	Low heat transfer rates, accumulation of dust in inlet grating further reducing heat transfer, dependent on wind direction and speed, low thermal conductivity and heat capacity of air, low mass flow rates of air, limited temperature reduction, not suited for roof-integrated solar panels	High initial cost for fans, ducts to handle large mass flow rates, high electrical consumption, high maintenance costs, noisy system, difficult to integrate compared to natural air circulation system	Higher initial cost due to pumps, higher maintenance cost compared to forced air circulation, higher electricity consumption compared to forced air circulation, a lower lifetime compared to forced air circulation due to corrosion	Low heat transfer rates, dust accumulation on the inlet grating, dependent on the wind speed and direction	Higher PCM cost compared to natural and forced air circulation, some PCMs are toxic, some PCMs have fire safety issues, some PCMs are strongly corrosive, PCMs may have disposal problem after their life cycle is complete	Heat transfer depends on ambient conditions, active systems, require electricity, reliability issues, costly for PV cooling–no heat storage capacity, requires efficient heat removal from the warmer side for effective cooling

transfer augmentation are necessary.

3.1.2. Forced air circulation

The irregular and uncontrolled nature of natural air circulation cooling method can be rectified by using forced air circulation. To understand the same, Adhikari et al. [24] presented a simulation model of thermal and electrical performances of an air cooled photovoltaic/thermal (PV/T) system. On the basis of energy balances of various components of the system, a new type (TYPE 25) was developed in FORTRAN and was integrated into TRNSYS software. The numerical calculation was carried out using a model developed for the climate of city Milan, Italy (45° 27′N). A simple economic analysis was conducted to compute the cost of unit energy production (€/kW h) for PV/T systems. The probable use of thermal energy produced by the PV / T system was also discussed and a sensitivity analysis on the cost of unit energy production was shown. The general preparation of a PV array, with a cooling duct fitted behind it, is indicated in Fig. 4. Most of the absorbed solar incident solar radiation appears passively as heat, levitation of cell temperature and thus reducing the efficiency with which the active part transforms light into electricity [25].

The loss of peak output power is measured around 1/2% per °C temperature rise for mono-crystalline silicon cell arrays [26]. The raised temperature of the array causes heat to be lost from the front face into the atmosphere, and from the rear face into the cooling duct. Buoyancy due to the warming of the air in the duct induces a rising flow so that the inwardly-transmitted heat is also removed to the surroundings, thus dropping the temperature of the array and restoring some of its efficiency. The induced flow may be supported or opposed by pressure differences between the inlet and outlet openings of the duct, due to local wind effects at those points. Brinkworth [27] stated based on numerical simulation and experimental validation that the ratio of duct length L to its hydraulic depth D should be about 20. For wide ducts, this equates to a length-to-depth ratio of about 40, giving practical and suitable dimensions. It is further revealed that the application of that rule is not affected significantly by the slope of the array.

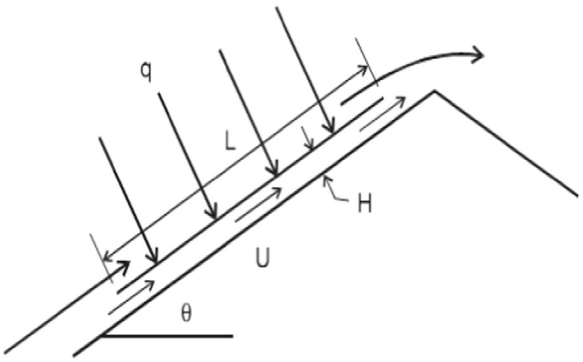


Fig. 4. General Arrangement of PV array with cooling duct [25].

Mittelman et al. [28] carried out a scale analysis and a numerical investigation to predict the passive air cooling rate of an open channel beneath PV panels. They showed that the effect of radiative heat transfer is greatest at low aspect ratios (or for short channels). System analysis also showed that the presence of the channel behind the PV panels can decline their temperature as much as 10–20 K which results in an absolute efficiency rise of 1–2%, depending on the channel geometry and the solar radiation. For example, for channel length L=3 m, channel spacing S =0.2 m, channel inclination angle relative to horizontal ϕ=30°, ambient and sky temperatures $T_o = T_{sky}$ =300 K, wind speed V =1 m/s and incident solar radiation G =900 W/m^2, the channel cooling rate, q_1, is 239 W/m^2 and the panel's surface temperature, T_1, is 338.7 K compared to 356.3 K without the channel. Because of the channel, the electrical production rate increases from 82 to 97 W_e/m^2. The rate of channel cooling was upsurges monotonically with the increase in channel spacing and decreases monotonically with the panel length. These results indicate that a passive cooling arrangement is a sound approach for regulating the temperature of rooftop integrated PV.

An experimental test setup was designed and fabricated by Teo et al. [29] to study the thermal and electrical performances of the PV/T air system. The experiment was planned to investigate the effects of temperature on the efficiency and power output of the PV system during operation. From the experimental result, it was shown that without active cooling, the operating temperature of the module

reached values as high as 68 °C and the electrical efficiency dropped significantly to 8.6%. By using a blower to cool the PV module, the operating temperature of the module could be sustained at 38 °C and the efficiency could also be kept at around 12.5%. Besides, an optimum flow rate was also found in this study. An air flow rate of 0.055 kg/s was sufficient to absorb the maximum amount of heat from the PV module. When the flow rate exceeded this value, the thermal and electrical energies were not affected.

Othman et al. [30] presented a study on different types of V-groove, honeycomb, and stainless steel wool heat exchangers. Heat exchangers were installed at the back side of a horizontal PV module. The system was tested at an irradiance of 828 W/m^2 with the mass flow rate ranging from 0.02 kg/s to 0.13 kg/s. It was perceived that the maximum thermal efficiency of the system with at a mass flow rate of 0.11 kg/s, with V-groove is 71%, stainless steel wool is 86% and honeycomb is 87% and with similar mass flow rate the electrical efficiency of the systems is 7.04%, 6.88%, and 7.13%, respectively and it was found that honeycomb design is the best effective design as a heat exchanger.

Farshchimonfared et al. [31] optimised the channel depth for the air mass flow rate per unit collector area and the air circulation duct diameter considering the system performance. They carried out numerical modelling of PV/T air collectors having several collector areas (A_c =10, 15, 25 and 30 m^2) and different length to width ratios (L/W =0.5, 1, 1.5 and 2) connected to an air distribution system. They showed that optimum depth (D_{opt}) value of PV/T air collectors varies between 0.09 and 0.026 m and the optimum air circulation duct varies in the range of 0.3 and 0.5 m. The optimum depth rises as the collector L/W ratio and the collector area (A_c) increase. Valeh-e-Sheyda et al. [32] carried out a study of two-phase air and water flows in a small hybrid micro-channel solar cell. An array of rectangular microchannels with a hydraulic diameter of 0.667 mm was used for conducting experiments. The velocity of air was in the range of 0–3.27 m/s while water flow rate was 0.04 m/s. The effect of two-phase operational fluid on PV cell cooling was compared with single-phase flow. Furthermore, the possibility of heat removal augmentation in PV/T panel was studied. The gained data showed the proposed hybrid system could considerably increase the output power of PV solar cells.

3.2. Water cooling

Fraisse et al. [33] presented a numerical study using software *TRNSYS*. It is based on connecting elementary modules called *TYPE*, which are either constituent of the studied system such as storage tanks or particular functions, for example, the weather data reader which permits connecting the selected area weather conditions to the system. They used three types of models, Global model, Thermal model of PV/T with a glazed cover (*TYPE* 105) and Electrical model of a photovoltaic module (*TYPE* 106). It was shown that the hybrid solar collectors represent a promising technology. It is mandatory that both photovoltaic and thermal field maintain a strong partnership in order to conceive a unique industrial constituent that is incorporated with the thermal absorber and the photovoltaic cells. Dubey et al. [34] carried a study on a photovoltaic cell which was attached partially or fully at its lower side to flat plate collectors. A detailed analysis of energy, exergy and electrical energy was presented and it was concluded that the partially covered collectors are beneficial in terms of annualized uniform cost (Rs. /kW h) if the primary requirement of user is thermal energy yield and fully covered collectors are beneficial when the primary requirement is electrical energy yield. This type of configurations is very useful in urban and rural areas, where hot water and electricity are required simultaneously. The annual prediction of yields obtained in terms of exergy, thermal energy and electrical energy for five different cities of India (Mumbai, New Delhi, Srinagar, Bangalore, and Jodhpur) showed that the Jodhpur had very high potential for installation of these water collectors.

Zhu et al. [35] presented an experimental study on water cooled PV module and concluded that the temperature of water-cooled PV module is lower than that of the air-cooled PV module. Therefore, the air-cooled module generates lower electrical power than the air-cooled one. The flow rate induces an irrelevant effect on the electrical efficiency of water-cooled PV modules. However, the thermal efficiency of water-cooled modules increases significantly with the flow rate. The PV/T collector using amorphous silicon PV cells and an active-cooling attachment can reach high overall energy conversion efficiency. Mishra and Tiwari [36] presented the study of the thermal energy, exergy, and electrical energy analysis of a Hybrid Photovoltaic Thermal (HPVT) Solar Water Heater. The study had been carried out by considering four different weather conditions for five cities (New Delhi, Jodhpur, Bangalore, Mumbai, and Srinagar) of India. Jodhpur has the maximum annual gain for the Jodhpur city and minimum for the Srinagar. The percentage variation between these three cities and Srinagar is 12%, 15.2%, and 10.4% for collectors partially covered by PV modules, and 10.2%, 14%, and 9.3% for collectors fully covered by PV modules, respectively.

Moharram et al. [37] presented an experimental and numerical study on cooling of PV cell using the least amount of water and energy. A cooling system was developed based on spraying of water on PV panels once in a day. Models of heating and cooling were validated experimentally. It was concluded that it is possible to cool and clean the PV panels using the proposed cooling system in hot and dusty regions. The PV panels produce the highest output energy if cooling of the panels starts when the temperature of the PV panels reaches the maximum permissible temperature, i.e., 45 °C. Haurant et al. [38] presented and evaluated a dynamic 3D numerical model for a high efficiency glazed flat plate PV–T collector. The model was centered on energy balance for each constituent control volume, joint with an interpretation of the fluid flow in terms of conductive and convective heat transfer equations. The model was validated under steady state and dynamic conditions, using data obtained under controlled laboratory test conditions. The model was found to truly reproduce the difference in outlet temperatures for a timescale of 1 min, with a precision of 1–2 °C for the most part of the order. However, inaccuracies as large as 6–7 °C was witnessed during brief peaks in temperature overlapping with sudden changes in flow rate.

3.3. Heat pipe cooling

Gang et al. [39] presented a numerical study on Heat-pipe photovoltaic/thermal (HP-PV/T) systems used in three typical climate areas of China namely, Hong Kong, Lhasa, and Beijing. The conclusions were drawn that the useful thermal and electrical energy obtained by HP-PV/T systems primarily depend on the available solar radiation in the area. It was shown that with the system equipped with auxiliary heating equipment in Hong Kong, Lhasa, and Beijing, the annual thermal energy is 1665.05–1872.22 MJ/m^2, 2939.67–3328.25 MJ/m^2, and 2111.07–2352.95 MJ/m^2, respectively; the annual electrical energy produced is 261.32–264.98 MJ/m^2, 462.14–466.1 MJ/ m^2, and 322.84–328.15 MJ/m^2, respectively. For the HP-PV/T system without auxiliary heating equipment, when ratio of hot-water load per unit collecting area is equal to 64.5 kg/m^2, in Hong Kong, there are 172 days a year where the hot water can be heated to more than 45 °C by means of solar energy whereas, in Lhasa and Beijing, there are 178 days and 158 days respectively. Zhang et al. [40] carried out a study on a photovoltaic/loop-heat-pipe (PV/LHP) heat pump system for potential use in space heating or hot water generation. They designed a numerical model that was able to predict the system's performance with a reasonable accuracy (mean errors no more than 9%). It was found that the average water temperature in the storage tank grew gradually from 14.38 to 54.29 °C over the testing duration. The temperature of the PV layer was in the range of 19.71– 52.64 °C during the daytime operation while its mean value was 40.66 °C. Wu

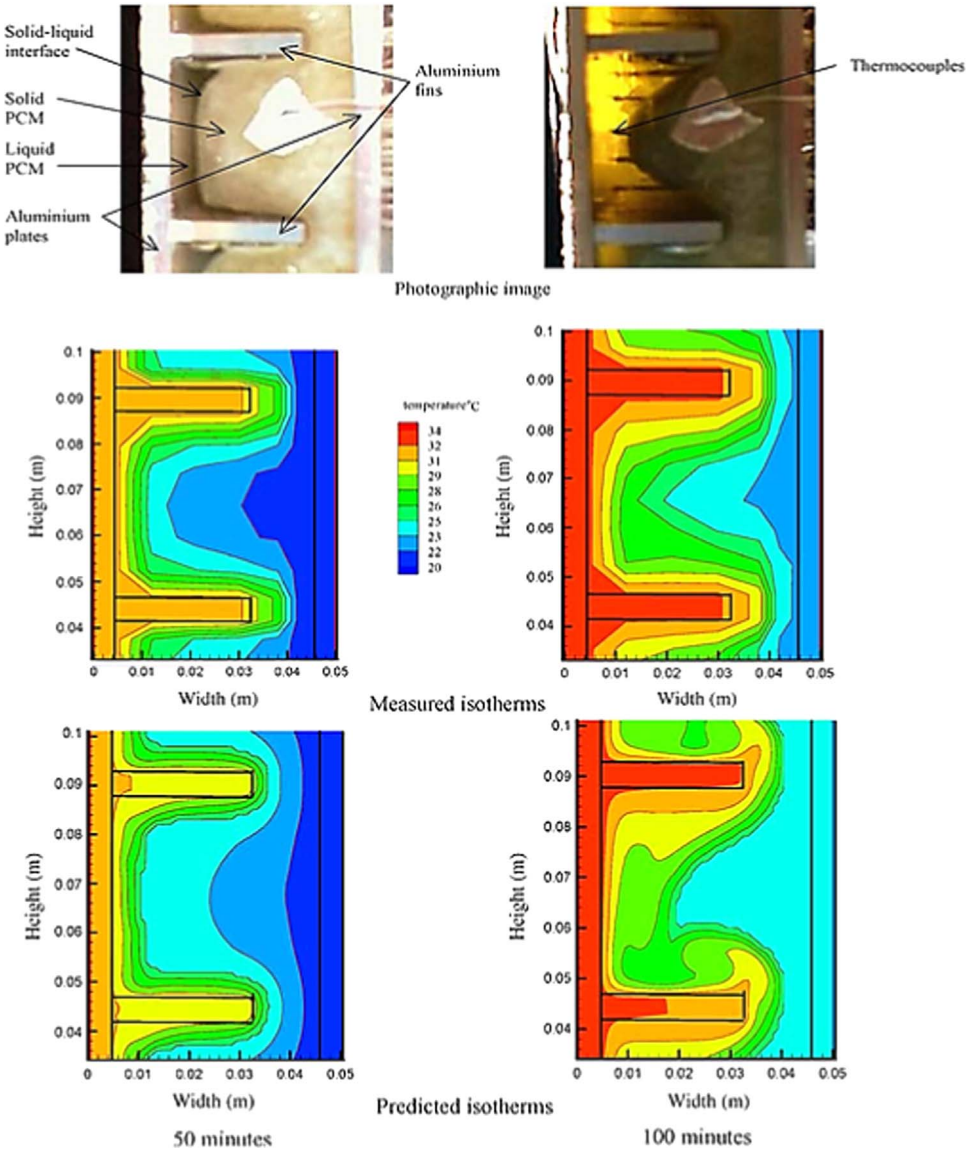


Fig. 5. Photovoltaic image of melt front and measured and predicted isotherms for a PV/PCM system with fins after 50 and 100 min with average insolation intensity at the front surface of 750 Wm^{-2} [42].

et al. [41] carried out a theoretical investigation to predict the performance of a heat pipe PV/T hybrid system. Results showed that the overall thermal, electrical and exergy efficiencies of heat pipe PV/T hybrid system could reach up to 63.65%, 8.45%, and 10.26%, respectively, while the varying range of operating temperature for the solar cell in the solar PV panel was less than 2.5 °C.

3.4. Cooling with Phase Change Materials (PCMs)

Huang et al. [42] carried out a numerical simulation study at 1000 (W/m^2) insolation and an ambient temperature of 20 °C, for a system with two fins immersed in a PCM of melt temperature 32 °C and a PCM container depth of 20 mm. The temperature on the front surface was preserved under 36.4 °C for 80 min. They found that to maintain the temperature of the front surface of the PV/PCM system at 25 °C, the transition temperature of the PCM should be higher than the ambient temperature and lower than 25 °C. The enhancement in the thermal performance achieved by using metal fins in the PCM container was significant. The fins allow a more uniform temperature distribution within the cooled PV/PCM system. However, they also found that increased numbers of fins can limit the movement of molten PCM as shown in Fig. 5.

Huang et al. [43] presented an experimental study on the application of PCM in Photovoltaic cell. Two specific phase change materials and three different PV/ PCM systems were used to moderate the temperature rise of photovoltaics. The PCMs were solid–liquid RT25 and Granular GR40, in which RT25 was found to be more effective than Granular PCM GR40. They presented a detailed experimental study to control the temperature at higher power density. Cellura et al. [44] carried out a theoretical investigation of a PV-PCM system in hot climate using software Comsol Multiphysics. By means of this software, the thermal behaviour of the PV-PCM system was simulated, thus permitting an improved identification of the suitable thermo-physical parameters for the optimisation of the panel efficiency. Simulation results showed that the proper application of PCM mass allows improved efficiency of the photovoltaic cell. Huang et al. [45] carried out another study on regulating the temperature of PV panel by application of phase change materials. Five PCMs were selected with melting temperature range of 25 ± 4 °C, three insolation intensities and latent heat of fusion from 140 to 213 kJ/kg. The different PCMs were Paraffin wax (RT20), Eutectic mixture of capric–lauric acid (C–L), Eutectic mixture of capric-palmitic acid (C–P), pure salt hydrate

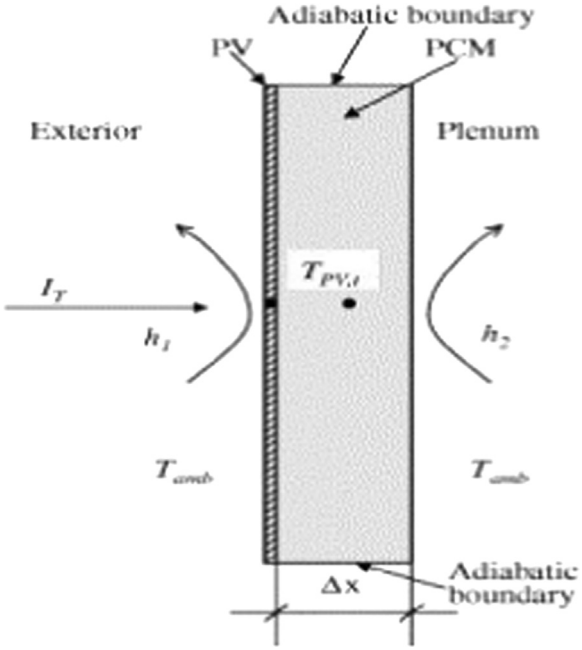


Fig. 6. Schematic diagram of heat transfer in PV/PCM system [45].

($CaCl_2{\cdot}6H_2O$), and Commercial blend (SP22). Comparing PCMs, the best results were obtained with C– P and $CaCl_2{\cdot}6H_2O$ that maintained a maximum of 18 °C temperature reductions at PV front surface for 30 min, while $CaCl_2$ maintained a 10 °C temperature reduction for the longest duration of 5 h at-1000 W/m^2 insolation onto the system.

A schematic diagram detailing the heat transfer in a PV/PCM system is presented in Fig. 6. The incident energy I_T increases the temperature of the PV/PCM as shown in Fig. 7.

An energy balance on a PV/PCM system for unit surface area can be written as [43]

$$\tau\alpha I_T \Delta t = \eta_c I_T \Delta t + U_L (T_{PV} - T_{amb})\Delta t + Q_S \tag{1}$$

where τ, α, Δt, η_c, U_L, T_{PV}, T_{amb} and Q_S respectively stand for the transmittance of the PV cover, the absorptance of the PV cell, the transition temperature of the PCM, the PV electrical conversion coefficient, the overall heat loss coefficient, the PV temperature, the ambient temperature, and the energy stored in the system. Assuming the difference in the specific heat of PCM in liquid and solid states is negligible and that the initial temperature is T_{amb}, the energy stored in the system is

$$Q_S = mc(T_{PV} - T_{amb}), \quad T_{amb} < T_{PV} < Tm \tag{2}$$

$$Q_S = mc(T_m - T_{amb}) + mH \quad T_m \le T_{PV} < Tm + \Delta T \tag{3}$$

$$Q_S = mc(T_m - T_{amb}) + mH + mc(T_{PV} - T_m), \ T_{PV} \ge Tm \tag{4}$$

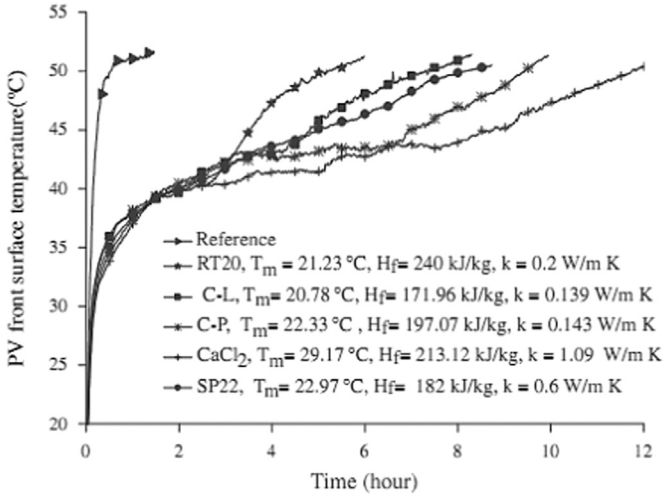


Fig. 7. Temperature evolution on the front surface of PV in system A, at an insolation of 750 W/m^2 and ambient temperature of 20 ± 1 °C [45].

Maiti et al. [46] carried out a study on a V-trough solar panel using a metal–wax composite phase changes matrix as shown in Fig. 8. Paraffin wax of 56–58 °C melting range was chosen as PCM and integrated at the rear of the module to absorb the excess heat. Two types of experiments were performed, indoor and outdoor. Using a 0.06 m thick layer of PCM, the module temperature in the indoor experiment could be maintained at 65–68 °C for 3 h whereas in the absence of PCM the temperature rose further than 90 °C within 15 min. In outside studies, the module temperature in V-trough could be reduced from 78 °C to 62 °C with the PCM assembly and operation could be sustained during the day. By means of the V-trough PV-PCM system, the output power of the day could be increased 1.55 times with self-regulation of temperature.

Huang [47] carried out a numerical study on an improved PV/PCM system with two PCMs with different phase transition temperatures for enhancing the heat regulation in building integrated photovoltaic. It was concluded that the use of two PCMs better helped in maintaining the PV panel close to its operating temperature, in comparison to a system with single PCM.

The governing equations used for PV/PCM system [47] include the momentum conservation equations (Navier–Stokes equation) in a Cartesian coordinate system:

$$\rho\frac{\overrightarrow{\partial u}}{\partial t} + \rho\vec{u}\frac{\partial\vec{u}}{\partial x} + \rho\vec{v}\frac{\partial\vec{u}}{\partial y} = \mu\nabla^2\vec{u} - \frac{\partial P}{\partial x} \tag{5}$$

$$\rho\frac{\overrightarrow{\partial v}}{\partial t} + \rho\vec{u}\frac{\partial\vec{v}}{\partial x} + \rho\vec{v}\frac{\partial\vec{v}}{\partial y} = \mu\nabla^2\vec{v} - \frac{\partial P}{\partial x} - \rho\vec{g} \tag{6}$$

where dynamic viscosity $\mu = \mu_S = \infty$ and density $\rho = \rho_S$ in solid regions of the PCM, and $\mu = \mu_L$, $\rho = \rho_L$ in liquid regions of the PCM. The momentum equations assume that the volume change in the liquid state is negligible in geometry but that it is reflected in the buoyancy driving force. The thermal expansion coefficient, β, is introduced into the momentum equation to include the buoyancy force term, according to the Boussinesq approximation:

$$\rho = \rho_0[1 - \beta(T - T_m)] \tag{7}$$

where ρ_o is the reference density and Tm is the melting or freezing temperature. Substituting ρ into the last term in Eq. (5) yields

$$-\rho g = \rho_0 g[\beta(T - T_m) - 1] \tag{8}$$

The energy equation for the melt PCM:

$$\rho_L C_L\frac{\partial T}{\partial x} + \frac{\partial}{\partial x}\left(\rho_L C_L\vec{u}T - k_L\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(\rho_L C_L\vec{v}T - k_L\frac{\partial T}{\partial y}\right) = 0 \tag{9}$$

The energy equation for solid parts:

$$\rho_S C_S\frac{\partial T}{\partial t} + \nabla(k_S\nabla T) = 0 \tag{10}$$

where the same equations hold good for all the frame and cell wall material and PCMs by including suitable k, ρ, and C_P values. The immediate continuity of heat flux and temperature at the interfaces of frame and cell with PCMs are preserved.

In the exterior front boundary, where the PV/PCM system is exposed to solar emission, the boundary condition is

$$k\frac{\partial T}{\partial x}\bigg|_{x=0} = S + h_0(T_{amb} - T_{x=0}) \tag{11}$$

In the exterior back layer of the PV/PCM system x=L, the boundary

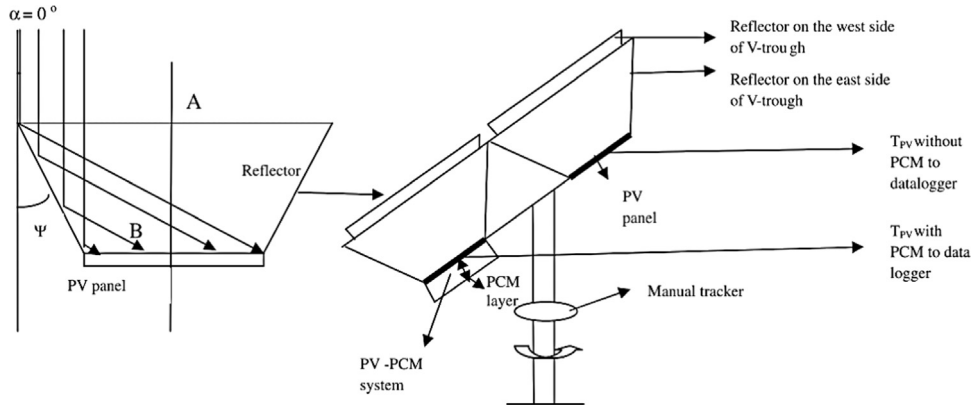


Fig. 8. Outdoor experimental set up for the PV module without PCM and PV-PCM system in a manually tracked V-trough [46].

condition is

$$k\frac{\partial T}{\partial x}\bigg|_{x=L} = S + h_L(T_{x=L} - T_{amb}) \tag{12}$$

where h_o and h_L are the heat transfer coefficients for the front and back surfaces of the PV/PCM system to the surroundings. S is the heat transferred by the PV/PCM system on the front surface from the incident solar energy. In the case of constant specific heat capacities for each phase, the temperature field can be defined as:

$$T = \begin{cases} E/C_s & T < T_m \quad SolidPhase \\ T_m & 0 < E < H \quad T > T_m \quad MeltZone \\ T_m + (E-H)/C_L & E \geq H \quad T > T_m \quad LiquidPhase \end{cases} \tag{13}$$

where E is the thermal energy and H the latent thermal energy.

Ho et al. [48] carried out a parametric simulation on a building integrated photovoltaic with microencapsulated phase change materials. The conclusion was drawn that the application of microencapsulated phase change materials (MEPCM) increases the minimum efficiency by approximately 0.13% in summer and 0.42% in winter condition when aspect ratio (A_m) of panel's width to panel's height is 0.277 and melting temperature of PCM (T_M) is 26 °C. In the normal condition, use of the MEPCM layer with A_m =0.277 and 1 increases and decreases the minimum efficiency by approximately 0.09% and 0.18%, respectively. These results suggested that there is an optimum aspect ratio lying between 0.277 and 1 for the normal case. The inclusion of the T_M =26 °C and 34◦C MEPCM layer increases the minimum efficiency by approximately 0.09% and 0.12%, respectively. Half of the T_M =34 °C layer does not melt after the 10 h experimental time, but for T_M =26 °C, almost all of the MEPCM melts.

Ciulla et al. [49] used PCM to enhance power production of photovoltaic cells. Finite difference method was used to predict the behaviour of PV-PCM systems. The results matched with experimental data obtained from a test facility. Biwole et al. [50] presented a computational fluid dynamics (CFD) modelling of PV system integrated with impure phase change materials. The numerical model presented in Fig. 9 was used to carry out a parametric study. All the simulations were conducted using ambient temperature =20 °C, external and internal heat transfer coefficients h_e =10 W/m^2 K, h_i =5 W/m^2 K respectively, constant solar radiation E =1000 W/m^2 and half melting temperature range ΔT=0.5 K. Different finned systems and PCM container geometry were simulated during the parametric study. The simulated SP/PCM systems allowed maintaining the panel's temperature under 40 °C during 80 min of constant exposure to a radiation of 1000 W/m^2. The same temperature was reached by the panel after only 5 min without the PCM.

Aelenei et al. [51] conducted an experimental and numerical simulation for the study of the performance of PV/PCM system at elevated temperature. Experimental and numerical results showed

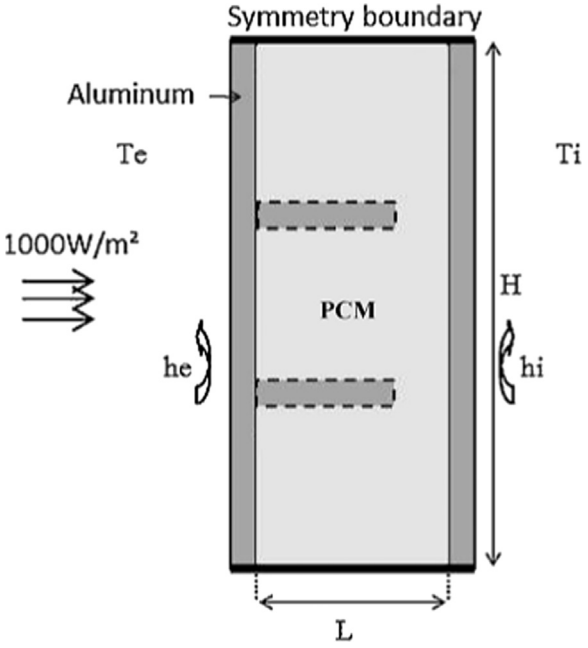


Fig. 9. Main elements of the numerical model [50].

relatively good agreement with each other. The maximum electrical and thermal efficiency of this BIPV-PCM could reach 10% and 12% respectively. Hasan et al. [52] fabricated a test set up and performed an experimental study of three 65 W polycrystalline EVA encapsulated PV panels of dimensions 771 mm×665 mm as shown in Fig. 10. It was concluded that life of PV panels increases due to reduced operating temperature and increased power density is gained for building integrated PV by the application of PCM at the back side of the panel.

Ho et al. [53] carried out a study on PV panel attached with a water-saturated MEPCM layer at its back side. Numerical simulations were used to analyse the capabilities of this module for controlling the PV panel temperature and power generation efficiency during the daytime. The highest temperature, average generation efficiency, and daily unit area generation output of the untreated PV was 32.8 °C, 19.46%, and 4646 kJ/m^2, whereas, for treated PV panels, it was 31.0 °C,19.61% and 4743.29 kJ/m^2 respectively.

Kibra et al. [54] developed thermal model to predict the thermal behaviour building integrated photovoltaic with phase change materials. Finite difference scheme was applied to discretize the energy balance equation while fully implicit scheme was applied to discretize the heat balance in the PCM module. Three different PCM of different melting temperatures were investigated. The numerical results were validated with the existing experimental studies. This study showed

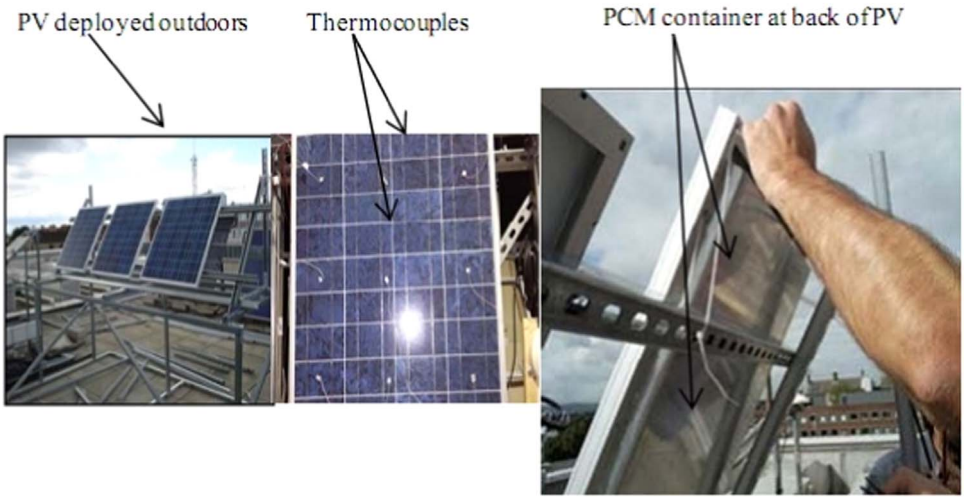


Fig. 10. Photographs of the experimental setup consisting of PV deployed outdoors at latitude angle of the selected sites, thermocouples installed at PV front surface and the PCM container integrated at the back of the PV [52].

Table 3
List of PCM which have been used/ can be used for regulating the temperature of the solar photovoltaic cell.

Phase change materials	Melting point (° C)	Latent heat (kJ/kg)	Density (kg/m^3)		Thermal conductivity (W/mK)		Specific heat (kJ/kg K)		Type
			Solid	Liquid	Solid	Liquid	Solid	Liquid	
RT25[55]	26.6	232	785	749	0.19	0.18	1.413	1797.6	Organic
Paraffin (PCM_1)[56]	26–28	179	870	750	0.2	–	1.8	2.4	Organic
Paraffin (PCM_2)[56]	31	169	870	760	0.2	–	1.8	2.4	Organic
Paraffin (PCM_3)[56]	28–32	157	880	760	0.2	–	1.8	2.4	Organic
Paraffin wax (RT20)[56]	25.73	140.3	880	770	0.2	–	1.8	2.4	Organic
Eutectic mixture of capric-lauric acid (C–L)[57,58]	24.66	171.98	880	863	0.139	–	–	–	Eutectic
Eutectic mixture of capric–palmitic acid (C–P)[57]	26.4	196.07	883	840	0.143	–	–	–	Eutectic
Pure salt hydrate ($CaCl_2{\cdot}6H_2O$)[59,60]	29.66	213.12	1710	–	1.09	–	1.4	–	Inorganic
Commercial blend (SP22)[61]	24.6	182	1490	1430	0.6		2.5		Inorganic
RT21[62]	21	134	760	840	0.2	–	–	–	Organic
RT27[62]	27	184	750	840	0.2	–	–	–	Organic
RT31[62]	29	196	770	890	0.2	–	–	–	Organic
RT60[62]	60	144	780	950	0.2	–	–	–	Organic
D-Lattic acid[63]	26	184	–	–	–	–	–	–	Organic
Methyl palmitate[63]	29	205							Organic
Polyethylene glycol 600($H(OC_2H_2)n{\cdot}OH$)[63]	20–25	146	–	–	–	–	–	–	Inorganic
$Mn(NO_3)_2{\cdot}6H_2O$[63]	25.5	148	–	–	–	–	–	–	Inorganic
$FeBr_3{\cdot}6H_2O$[63]	27	105	–	–	–	–	–	–	Inorganic
$CaCl_2{\cdot}12H_2O$[63]	29.8	174	–	–	–	–	–	–	Inorganic
$C_{14}H_{28}O_2 + C_{10}H_{20}O_2$[63]	24	147.7	–	–	–	–	–	–	Eutectic
$CaCl_2 + MgCl_2{\cdot}6H_2O$[63]	25	95	–	–	–	–	–	–	Eutectic
$CH_3CONH_2 + NH_2CONH_2$[63]	27	163	–	–	–	–	–	–	Eutectic
Triethylolethane + urea[63]	29.8	218							Eutectic

that PCM could be quite effective for limiting the temperature rise in the PV devices and hence increasing the thermal performance up to 5%.

The list of various PCMs and their thermal properties that can be used for regulating the temperature of PV panels has been listed in Table 3.

3.5. Heat transfer enhancement of phase change materials

Most PCMs have low thermal conductivity, which slows their heat charging and discharging rate and limits their application. Several studies have been conducted for the heat transfer enhancement of phase change materials included in tubes of different configurations [64–75], bubble agitation [73], insertion of a metal matrix into the PCM [65,66], using PCM dispersed with high conductivity particles [76], micro-encapsulation of the PCM [77,78] or shell and tube (multi-tubes) [79,80]. Velraj et al. [81] evaluated the enhancement of the heat transfer using the effective thermal conductivity taken from a two-dimensional enthalpy-temperature governing equation which assumed no variation of temperature and thermal conductivity in the axial direction. Results were presented graphically using temperature–time.

3.5.1. Encapsulation and use of metallic fins

Naturally, the insertion of highly conductive materials to form a composite of the PCM and thermal conductivity promoter is a consistent solution. The insertion of metal fins, foams, wools into the PCM has long been practiced. Determining proper configurations of these fixed, non-moving structures and their interactions with conduction, convection and phase change heat transport mechanisms pose challenging issues. A review carried out by Fan and Khodadadi [82]

Table 4
Summary of research work performed so far for thermal regulation of PV panels.

Author	Year	Configuration	PV type	Technique (Exp. Or Simulation)	Electrical efficiency of PV cell	Thermal efficiency	PV cell temperature	Conclusion
Air cooling								
Adhikari et al. [24]	2003	Forced Air circulation	Multi-crystalline silicon	Simulation	–	30–40%	Increment of 5–15 °C of outlet air Temperature	Efficiency of PV cell Increased
B.J. Brinkworth et al. [26]	2006	Forced Air circulation	–	Theoretical	–	–	–	L/D=20
Brinkworth [27]	2006	Forced Air circulation	–	Theoretical	–	–	–	Optimum geometry, $(L/D)_{opt}=2(Re_D)^{0.24}$
Mittelman et al. [28]	2009	Forced Air circulation	Polycrystalline silicon	Numerical simulation	–	–	65 °C	Absolute efficiency gain of 1–2%
Teo et al. [29]	2012	Forced Air circulation	Polycrystalline	Experimental and Simulation	12.5%	–	38 °C	Temperature reduced 40° C and increment in efficiency 3.9%
Othman et al. [30]	2013	Heat Exchanger	Mono-crystalline	Experimental	7.13%	87%	$T_{out}–T_{in}$=8 °C	Honeycomb heat exchanger is the most efficient
Farshchimonfared et at. [31]	2015	Forced circulation of air with different collector area	–	Experimental	–	–	Difference in temperature was 10 °C	Optimisation of PV/T collector area with aspect ratio
Sheyda et al. [32]	2013	Forced air circulation with microchannel	Polycrystalline silicon	Experimental	–	–	–	Maximum output power of PV cell was about 38%
Water								
Fraiss et al. [33]	2007	Water circulation	PV/T with a glazed cover	Numerical modelling	6.8%	–	35 °C	–
Dubey et al. [34]	2009	Forced circulation of water and air	Thermal collector partially covered with PV	Theoretical	8.5–10.5%	38%	40–90 °C	–
Zhu et al. [35]	2012	Forced circulation	Si PV cell with air and Water	Experimental	–	Varies from 37.5% to 62.1% according to flow rate	33.1 °C	Overall efficiency gain of 3.4%
Mishra and Tiwari [36]	2012	Forced circulation of water and air	PVT with fully and partially covered	Experimental	12%	–	–	Total annual gain is maximum for the Jodhpur city
Moharram et al. [37]	2013	Forced circulation of water	Monocrystalline silicon	Experimental	12.5%	–	45 °C	Panel temperature was reduced by 10 °C
Haurant et al. [38]	2015	A glass covered water based PV–T prototype	–	Numerical 3D dynamic model	8–11%	–	–	–
Heat pipe cooling								
Gang et al. [39]	1999	Heat pipe PV/T system	–	Experimental and Theoretical	_	_	_	When Mw/Ac is equal to 64.5 kg/m^2, the annual solar thermal contribution of the HP-PV/T system in Hong Kong, Lhasa, and Beijing are 68.5%, 80.5%, and 64.7%, respectively
Zhang et al. [40]	2014	Heat pipe PV/T system		Theoretical computer simulation, experimental verification	–	–	–	average water temperature in the storage tank grew gradually from 14.38 to 54.29 °C over the testing duration
Wu et al. [41]	2011	Heat pipe PV/T system	–	theoretical model	–	–	–	exergy efficiencies of heat pipe PV/T hybrid system could reach up to 63.65%, 8.45% and 10.26%, respectively
PCM								
Huang et al. [42]	2004	Phase change materials RT25 and Paraffin wax	PV/PCM system with metal fins	Experiments and numerical simulations	–	–	34 °C	Fins enable more uniform heat distribution
Huang et al. [43]	2006	Phase change materials RT25 and GR40	PV/PCM with three type of fin configuration	Experimental and numerical simulations	–	–	29 °C	RT25 gives better performance as compared to other
Cellura et al. [44]	2008	Three type of PCM (paraffins) used	PV/PCM	Simulations	–	–	40–50 °C	–

(continued on next page)

Table 4 *(continued)*

Author	Year	Configuration	PV type	Technique (Exp. Or Simulation)	Electrical efficiency of PV cell	Thermal efficiency	PV cell temperature	Conclusion
Huang et al. [45]	2010	Five type of PCM with three insolation intensities	PV/PCM	Experimental	–	–	18 °C temperature reduction in PV front surface for 30 min	$CaCl_2$ is best as compare to other
Maiti et al. [46]	2011	Paraffin wax was used as phase change materials	PV/PCM with V-trough	Experimental	5.40%	–	65–68 °C	Temperature of PV was reduced and increased cell efficiency
Huang [47]	2011	Four type of PCM RT21, RT27, RT31, and RT60 was used	PV/PCM with triangular and half circular aluminium metal cells at rear of PV	Parametric analysis	–	–	–	RT27–RT21 achieves the highest temperature reduction
Ho et al. [48]	2012	MEPCM applied at rear end of PV	BIPV with microencapsulated	Numerical	–	–	Melting temperature of PCM was 26–36 °C	Efficiency increased up to 0.13–0.40%
Ciulla et al. [49]	2012	Finite difference model	–	Mathematical modelling	–	–	Transition phase of PCM 26–28 °C	–
Biwole et al. [50]	2013	Numerical simulation with and without fins	PV/PCM	Detailed mathematical and numerical modelling	–	–	40 °C	Panel's temperature under 40 °C during 80 min
Aelenei et al. [51]	2014	BIPV-PCM installed on roof	PV -polycrystalline	Theoretical simulation	10%	12%	–	Overall system efficiency (electrical and thermal) of about 20%.
Hasan et al. [52]	2014	Experimental investigation of PV/PCM system at elevated temperature	65 W polycrystalline EVA encapsulated PV panels	Experimental	–	–	43 and 51 °C	–
Ho et al. [53]	2015	Water surface floating PV with MEPCM	PV/PCM with floating on water	Numerical simulation			30–31 °C	The generated output of the treated PV increased by 2.1% relative to the untreated PV.
Kibria et al. [54]	2016	Numerical investigation	BIPV/PCM	Thermal modelling	–	–	–	Thermal performance increased up to 5%

shows that the utilisation of non-moving structures in PCMs for enhancing thermal conductivity is an age old and well established technique.

3.5.2. Addition of highly thermal conductive nanoparticles

Recognising the variety of enhancements/functionalities brought due to nanotechnology, another possibility to enhance thermal conductivity of PCMs is by introducing nanostructures (nanoparticles, nanotubes, nanofibers, etc.) into PCM. These composite Nano enhanced PCMs provide extra functionalities and enhancements not obtainable to the date, with thermal conductivity enhancement. In this regards, composites of solids and nanostructures to form structural materials had been widely investigated and dispersions of nanostructures in liquids (also termed nanofluids) have received widespread attention by the research community [83]. However, PCMs are unique materials which are utilised in demanding freeze/ thaw cycles. Thus, PCM is expected to perform in both liquid and solid phases and their variation using nano structures has been attempted over the past few years.

The modest model for forecasting the effective thermal conductivity (k) of double phase mixtures is the mixture rule that takes the following form Das et al. [83].

$$k_{eff} = (1 - \phi)k_c + \phi k_d \tag{14}$$

where subscripts *eff*, *c* and *d* stand for effective, continuous and discrete phases respectively, and ϕ is the volume fraction of the discrete phase. The effective thermal conductivity of nano structured phase change materials can be predicted using the Maxwell [84] relation which is applicable to the volume fraction of nanoparticles up to 10% [85].

$$k_{eff} = \frac{k_d + 2k_c - 2\phi(k_c - k_d)}{k_d + 2k_c + \phi(k_c - k_d)} \tag{15}$$

Based on abundant experimental data obtained by over 30 groups worldwide on identical samples, Buongiorno et al. [86] found that the measured thermal conductivity of Nanofluids (volume fraction less than 3%) was in good agreement with Maxwell's equation and the extended model proposed by Nan et al. [87], and no anomalous enhancement beyond the static part was observed.

3.6. Thermoelectric cooling

Thermoelectric cooling by Peltier effect is a popular but inefficient way to remove heat from high-power-dissipating electrical components. Thermoelectric cooling in electronics cooling applications is often favored over other means because thermoelectrics is silent, compact, reliable and durable. In addition, the cooling power can be modulated to maintain a fixed temperature. In those cases, Peltier elements are coupled with additional heat exchangers to remove the heat from their hot plate.

4. Conclusion

In this paper an extensive review has been carried out for various methodologies used for cooling PV panels to enhance their electrical efficiency. Techniques such as air cooling, water cooling, heat pipe cooling, thermoelectric cooling and cooling with phase change materials are discussed and compared for the regulating the temperature of

the photovoltaic cells. The summary of the various researchers conducted so far has been given in Table 4. It can be concluded from this study that cooling technologies are essential in the solar circuit, in order to take maximum advantage of the solar photovoltaic cell and control the temperature rise of photovoltaic. Among above discussed techniques cooling with phase change materials is the most promising and effective cooling technique for photovoltaic due to its higher energy density per unit volume. The current status of research in respect to this technology is abridged above, to recognising the finished work, unresolved problems, the potential and barriers to practical application as well.

However, an economic evaluation is required for the different thermal management techniques, to determine if the extra capital cost of additional arrangement is indeed outweighed by additional energy savings. Then the suitability, applicability, and impacts generated by the systems should be addressed at the environmental and social levels. It can be expected that these systems will have a relatively long payback period taking into account the initial cost of PV panels and the additional cost of the cooling arrangements. Providentially, if the thermal energy extracted from PV panels can be effectively used for space heating, ventilation, and hot water, the economic feasibility is expected to be enhanced. Additional work, therefore, should be conducted from the perspectives of life cycle assessment and economic estimation.

References

[1] J.J. Wysocki, P. Rappaport, Effect of temperature on photovoltaic solar energy conversion, J. Appl. Phys. 31 (1960) 571–578.

[2] Y. Charabi, A. Gastli, Integration of temperature and dust effects in siting large PV power plant in hot arid area, Renew. Energy 57 (2013) 635–644. http://dx.doi.org/10.1016/j.renene.2013.02.031.

[3] S. Dubey, J.N. Sarvaiya, B. Seshadri, Temperature Dependent Photovoltaic (PV) efficiency and its effect on PV production in the world – a review, Energy Procedia 33 (2013) 311–321. http://dx.doi.org/10.1016/j.egypro.2013.05.072.

[4] J.K. Kaldellis, M. Kapsali, K. a. Kavadias, Temperature and wind speed impact on the efficiency of PV installations. Experience obtained from outdoor measurements in Greece, Renew. Energy 66 (2014) 612–624. http://dx.doi.org/10.1016/j.renene.2013.12.041.

[5] M.E. Ya'acob, H. Hizam, a.M. Bakri, M.A.M. Radzi, T. Khatib, a.H.M. a. Rahim, Performance test conditions for direct temperature elements of multiple PV array configurations in Malaysia, Energy Proc. 61 (2014) 2387–2390. http://dx.doi.org/10.1016/j.egypro.2014.12.011.

[6] W.G.J. Van Helden, R.J.C. van Zolingen, H.A. Zondag, PV thermal systems: PV panels supplying renewable electricity and heat, Prog. Photovolt.: Res. Appl. 12 (2004) 415–426.

[7] A. Luque, S. Hegedus, Handbook of Photovoltaic Science and Engineering, John Wiley & Sons, 2011.

[8] S.R.Kurtz, Opportunities and challenges for development of a mature concentrating photovoltaic power industry, 2009.

[9] E. Radziemska, The effect of temperature on the power drop in crystalline silicon solar cells, Renew. Energy 28 (2003) 1–12.

[10] B. Du, E. Hu, M. Kolhe, Performance analysis of water cooled concentrated photovoltaic (CPV) system, Renew. Sustain. Energy Rev. 16 (2012) 6732–6736.

[11] S. Chatterjee, G. TamizhMani, BAPV arrays: Side-by-side comparison with and without fan cooling, in: Proceedings of the 38th IEEE Photovoltaic Specialists Conference (PVSC), 2012, IEEE, 2012: pp. 537–542

[12] M.Simon, On the evaluation of spectral effects on photovoltaic modules performance parameters and hotspots in solar cells, 2009.

[13] B.J. Huang, P.E. Yang, Y.P. Lin, B.Y. Lin, H.J. Chen, R.C. Lai, et al., Solar cell junction temperature measurement of PV module, Sol. Energy 85 (2011) 388–392.

[14] V.M.V.M. Andreev, V.A. Grilikhes, V.D. Rumiantsev, Photovoltaic Conversion of Concentrated Sunlight, Wiley, 1997.

[15] D.L. King, P.E. Eckert, Characterizing (rating) the performance of large photovoltaic arrays for all operating conditions, in: Proceedings of the Photovoltaic Specialists Conference, 1996, Conference Record of the Twenty Fifth IEEE, IEEE, 1996: pp. 1385–1388.

[16] B. Moshfegh, M. Sandberg, Flow and heat transfer in the air gap behind photovoltaic panels, Renew. Sustain. Energy Rev. 2 (1998) 287–301.

[17] J.A. Del Cueto, Comparison of energy production and performance from flat-plate photovoltaic module technologies deployed at fixed tilt, in: Proceedings of the Photovoltaic Specialists Conference, 2002. Conference Record of the Twenty-Ninth IEEE, IEEE, 2002: pp. 1523–1526

[18] L. Yang, Q.H. Ye, A. Ebong, W.T. Song, G.J. Zhang, J.X. Wang, et al., High efficiency screen printed bifacial solar cells on monocrystalline CZ silicon, Prog. Photovolt.: Res. Appl. 19 (2011) 275–279.

[19] B.P. Jelle, C. Breivik, H.D. Røkenes, Building integrated photovoltaic products: a state-of-the-art review and future research opportunities, Sol. Energy Mater. Sol. Cells 100 (2012) 69–96.

[20] E.C. Kern Jr, M.C. Russell, Combined photovoltaic and thermal hybrid collector systems, Massachusetts Inst. of Tech. Lexington (USA). Lincoln Lab., 1978.

[21] X. Zhang, X. Zhao, S. Smith, J. Xu, X. Yu, Review of R & D progress and practical application of the solar photovoltaic / thermal (PV / T) technologies, Renew. Sustain. Energy Rev. 16 (2012) 599–617. http://dx.doi.org/10.1016/j.rser.2011.08.026.

[22] T.T. Chow, A review on photovoltaic/thermal hybrid solar technology, Appl. Energy 87 (2010) 365–379. http://dx.doi.org/10.1016/j.apenergy.2009.06.037.

[23] K. Moradi, M. Ali Ebadian, C.X. Lin, A review of PV/T technologies: effects of control parameters, Int. J. Heat. Mass Transf. 64 (2013) 483–500. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2013.04.044.

[24] R.S.Adhikari, F.Butera, P.Caputo, P.Oliaro, N.Aste, Thermal and electrical performances of a new kind air cooled photovoltaic thermal system for building application, in: Proceedings of the ISES Solar World Congress, Goteborg, 2003.

[25] B.J.J. Brinkworth, M. Sandberg, Design procedure for cooling ducts to minimise efficiency loss due to temperature rise in PV arrays, Sol. Energy 80 (2006) 89–103. http://dx.doi.org/10.1016/j.solener.2005.05.020.

[26] B. Brinkworth, B. Cross, R. Marshall, H. Yang, Thermal regulation of photovoltaic cladding, Sol. Energy 61 (1997) 169–178. http://dx.doi.org/10.1016/S0038-092X(97)00044-3.

[27] B.J. Brinkworth, Optimum depth for PV cooling ducts, Sol. Energy 80 (2006) 1131–1134. http://dx.doi.org/10.1016/j.solener.2005.09.006.

[28] G. Mittelman, A. Alshare, J.H. Davidson, A model and heat transfer correlation for rooftop integrated photovoltaics with a passive air cooling channel, Sol. Energy 83 (2009) 1150–1160. http://dx.doi.org/10.1016/j.solener.2009.01.015.

[29] H.G. Teo, P.S. Lee, M.N.A. Hawlader, An active cooling system for photovoltaic modules, Appl. Energy 90 (2012) 309–315. http://dx.doi.org/10.1016/j.apenergy.2011.01.017.

[30] M.Y.H. Othman, F. Hussain, K. Sopian, B. Yatim, H. Ruslan, K. Prestasi, et al., Performance study of air-based photovoltaic-thermal (PV/T) collector with different designs of heat exchanger, Sains Malays. 42 (2013) 1319–1325.

[31] M. Farshchimonfared, J.I.I. Bilbao, a.B.B. Sproul, Channel depth, air mass flow rate and air distribution duct diameter optimization of photovoltaic thermal (PV/T) air collectors linked to residential buildings, Renew. Energy 76 (2015) 27–35. http://dx.doi.org/10.1016/j.renene.2014.10.044.

[32] P. Valeh-E-Sheyda, M. Rahimi, E. Karimi, M. Asadi, Application of two-phase flow for cooling of hybrid microchannel PV cells: a comparative study, Energy Convers. Manag. 69 (2013) 122–130. http://dx.doi.org/10.1016/j.enconman.2013.01.029.

[33] G. Fraisse, C. Ménézo, K. Johannes, Energy performance of water hybrid PV/T collectors applied to combisystems of direct solar floor type, Sol. Energy 81 (2007) 1426–1438.

[34] S. Dubey, G.N. Tiwari, Analysis of PV/T flat plate water collectors connected in series, Sol. Energy 83 (2009) 1485–1498.

[35] Q. Zhu, L. Si, Electrical outputs and thermal outputs of water/air cooled amorphous-silicon photovoltaic modules, Adv. Biomed. Eng. 8 (2012) 83.

[36] R.K. Mishra, A. Tiwari, Study of Hybrid Photovoltaic Thermal (HPVT) solar water heater at constant collection temperature for indian climatic conditions, J. Fundam. Renew. Energy Appl. 2 (2012). http://dx.doi.org/10.4303/jfrea/R120310.

[37] K.A. Moharram, M.S. Abd-Elhady, H.A. Kandil, H. El-Sherif, Enhancing the performance of photovoltaic panels by water cooling, Ain Shams Eng. J. 4 (2013) 869–877. http://dx.doi.org/10.1016/j.asej.2013.03.005.

[38] H. Pierrick, M. Christophe, G. Leon, D. Patrick, Dynamic numerical model of a high efficiency PV–T collector integrated into a domestic hot water system, Sol. Energy 111 (2015) 68–81.

[39] P. Gang, F. Huide, J. Jie, C. Tin-Tai, Z. Tao, Annual analysis of heat pipe PV/T systems for domestic hot water and electricity production, Energy Convers. Manag. 56 (2012) 8–21. http://dx.doi.org/10.1016/j.enconman.2011.11.011.

[40] X. Zhang, X. Zhao, J. Shen, J. Xu, X. Yu, Dynamic performance of a novel solar photovoltaic/loop-heat-pipe heat pump system, Appl. Energy 114 (2014) 335–352. http://dx.doi.org/10.1016/j.apenergy.2013.09.063.

[41] S.Y. Wu, Q.L. Zhang, L. Xiao, F.H. Guo, A heat pipe photovoltaic/thermal (PV/T) hybrid system and its performance evaluation, Energy Build. 43 (2011) 3558–3567. http://dx.doi.org/10.1016/j.enbuild.2011.09.017.

[42] M.J. Huang, P.C. Eames, B. Norton, Thermal regulation of building-integrated photovoltaics using phase change materials, Int. J. Heat. Mass Transf. 47 (2004) 2715–2733. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2003.11.015.

[43] M.J.J. Huang, P.C.C. Eames, B. Norton, Phase change materials for limiting temperature rise in building integrated photovoltaics, Sol. Energy 80 (2006) 1121–1130. http://dx.doi.org/10.1016/j.solener.2005.10.006.

[44] M. Cellura, G. Ciulla, V. Lo Brano, A.Marvuglia, A Photovoltaic panel coupled with a phase changing material heat storage system in hot climates, PLEA 2008, in: Proceedings of the 25th Conference on Passive and Low Energy Architecture, Dublin, 22nd to 24th October 2008. (2008) 22–27

[45] A. Hasan, S.J.J. McCormack, M.J.J. Huang, B. Norton, Evaluation of phase change materials for thermal regulation enhancement of building integrated photovoltaics, Sol. Energy 84 (2010) 1601–1612. http://dx.doi.org/10.1016/j.solener.2010.06.010.

[46] S. Maiti, S. Banerjee, K. Vyas, P. Patel, P.K. Ghosh, Self regulation of photovoltaic module temperature in V-trough using a metal–wax composite phase change matrix, Sol. Energy 85 (2011) 1805–1816. http://dx.doi.org/10.1016/j.solener.2011.04.021.

[47] M. Jun Huang, The effect of using two PCMs on the thermal regulation performance of BIPV systems, Sol. Energy Mater. Sol. Cells 95 (2011) 957–963.

http://dx.doi.org/10.1016/j.solmat.2010.11.032.
[48] C.J. Ho, a.O. Tanuwijava, C.-M. Lai, Thermal and electrical performance of a BIPV integrated with a microencapsulated phase change material layer, Energy Build. 50 (2012) 331–338. http://dx.doi.org/10.1016/j.enbuild.2012.04.003.
[49] G. Ciulla, V. Lo Brano, M. Cellura, V. Franzitta, D. Milone, V. Lo Brano, et al., A finite difference model of a PV-PCM system, Energy Procedia 30 (2012) 198–206. http://dx.doi.org/10.1016/j.egypro.2012.11.024.
[50] P.H. Biwole, P. Eclache, F. Kuznik, Phase-change materials to improve solar panel's performance, Energy Build. 62 (2013) 59–67. http://dx.doi.org/10.1016/j.enbuild.2013.02.059.
[51] L. Aelenei, R. Pereira, H. Gon??alves, A. Athienitis, H. Gonçalves, A. Athienitis, Thermal performance of a hybrid BIPV-PCM: modeling, design and experimental investigation, Energy Procedia 48 (2014) 474–483. http://dx.doi.org/10.1016/j.egypro.2014.02.056.
[52] A. Hasan, S.J. Mccormack, M.J. Huang, B. Norton, Energy and cost saving of a photovoltaic-phase change materials (PV-PCM) system through temperature regulation and performance enhancement of photovoltaics, Energies 7 (2014) 1318–1331. http://dx.doi.org/10.3390/en7031318.
[53] C.J. Ho, W.-L. Chou, C.-M. Lai, Thermal and electrical performance of a water-surface floating PV integrated with a water-saturated MEPCM layer, Energy Convers. Manag. 89 (2015) 862–872.
[54] M.A. Kibria, R. Saidur, F.A. Al-Sulaiman, M.M.A. Aziz, Development of a thermal model for a hybrid photovoltaic module and phase change materials storage integrated in buildings, Sol. Energy 124 (2016) 114–123. http://dx.doi.org/10.1016/j.solener.2015.11.027.
[55] W. Lu, S.A. Tassou, Experimental study of the thermal characteristics of phase change slurries for active cooling 91 2012 366 374, ⟨http://dx.doi.org/10.1016/j.apenergy.2011.10.004⟩.
[56] B. He, V. Martin, F. Setterwall, Phase transition temperature ranges and storage density of paraffin wax Phase change materials, Energy 29 (2004) 1785–1804. http://dx.doi.org/10.1016/j.energy.2004.03.002.
[57] A. Sharma, A. Shukla, C.R. Chen, S. Dwivedi, Development of phase change materials for building applications, Energy Buildings 64 (2013) 403–407. http://dx.doi.org/10.1016/j.enbuild.2013.05.029.
[58] A. Sharma, a. Shukla, Thermal cycle test of binary mixtures of some fatty acids as phase change materials for building applications, Energy Build. 99 (2015) 196–203. http://dx.doi.org/10.1016/j.enbuild.2015.04.028.
[59] V. Kapsalis, D. Karamanis, Solar thermal energy storage and heat pumps with phase change materials, Appl. Therm. Eng. (2016). http://dx.doi.org/10.1016/j.applthermaleng.2016.01.071.
[60] L.F.F. Cabeza, A. Castell, C. Barreneche, A. De Gracia, A.I. Fern??ndez, A.I.I. Fernández, et al., Materials used as PCM in thermal energy storage in buildings: a review, Renew. Sustain. Energy Rev. 15 (2011) 1675–1695. http://dx.doi.org/10.1016/j.rser.2010.11.018.
[61] S. Sharma, A. Tahir, K.S.S. Reddy, T.K. Mallick, Performance enhancement of a Building-Integrated Concentrating Photovoltaic system using phase change material, Sol. Energy Mater. Sol. Cells 149 (2016) 29–39. http://dx.doi.org/10.1016/j.solmat.2015.12.035.
[62] Rubitherm GmbH, Available at ⟨Http://www.rubitherm.de⟩. (2016). ⟨http://www.rubitherm.de/⟩.
[63] A. Sharma, V.V. Tyagi, C.R. Chen, D. Buddhi, Review on thermal energy storage with phase change materials and applications, Renew. Sustain. Energy Rev. 13 (2009) 318–345. http://dx.doi.org/10.1016/j.rser.2007.10.005.
[64] R.M. Abdel-Wahed, J.W. Ramsey, E.M. Sparrow, Photographic study of melting about an embedded horizontal heating cylinder, Int. J. Heat. Mass Transf. 22 (1979) 171–173.
[65] K. Ermis, A. Erek, I. Dincer, Heat transfer analysis of phase change process in a finned-tube thermal energy storage system using artificial neural network, Int. J. Heat. Mass Transf. 50 (2007) 3163–3175.
[66] K.A.R. Ismail, C.L.F. Alves, M.S. Modesto, Numerical and experimental study on the solidification of PCM around a vertical axially finned isothermal cylinder, Appl. Therm. Eng. 21 (2001) 53–77.
[67] J.C. Choi, S.D. Kim, Heat-transfer characteristics of a latent Heat storage system using MgCl2·6H2O, Energy 17 (1992) 1153–1164.
[68] F.Agyenim, N.Hewitt, P.Eames, M.Smyth, Numerical and experimental development of medium temperature thermal energy storage (Erythritol) system for the hot side of LiBr/H2O air conditioning applications, in: World Renewable Energy Congress, 2008.
[69] B. Horbaniuc, G. Dumitrascu, A. Popescu, Mathematical models for the study of solidification within a longitudinally finned heat pipe latent heat thermal storage system, Energy Convers. Manag. 40 (1999) 1765–1774.
[70] E.M. Sparrow, E.D. Larson, J.W. Ramsey, Freezing on a finned tube for either conduction-controlled or natural-convection-controlled heat transfer, Int. J. Heat. Mass Transf. 24 (1981) 273–284.
[71] K. Sasaguchi, H. Takeo, Effect of the orientation of a finned surface on the melting of frozen porous media, Int. J. Heat. Mass Transf. 37 (1994) 13–26.
[72] Y. Zhang, A. Faghri, Heat transfer enhancement in latent Heat thermal energy storage system by using the internally finned tube, Int. J. Heat. Mass Transf. 39 (1996) 3165–3173.
[73] R. Velraj, R.V. Seeniraj, B. Hafner, C. Faber, K. Schwarzer, Experimental analysis and numerical modelling of inward solidification on a finned vertical tube for a latent heat storage unit, Sol. Energy 60 (1997) 281–290.
[74] J.P. Trelles, J.J. Dufly, Numerical simulation of porous latent heat thermal energy storage for thermoelectric cooling, Appl. Therm. Eng. 23 (2003) 1647–1664.
[75] C.J. Hoogendoorn, G.C.J. Bart, Performance and modelling of latent heat stores, Sol. Energy 48 (1992) 53–58.
[76] E.-B.S. Mettawee, G.M.R. Assassa, Thermal conductivity enhancement in a latent heat storage system, Sol. Energy 81 (2007) 839–845.
[77] P.W. Griffiths, P.C. Eames, Performance of chilled ceiling panels using phase change material slurries as the heat transport medium 27 (2007) 1756–1760. http://dx.doi.org/10.1016/j.applthermaleng.2006.07.009.
[78] K.M. Hawlader MNA, M.S. Uddin, Microencapsulated PCM thermal- energy storage system, Appl. Energy 74 (2003) 195–202.
[79] F. Agyenim, P. Eames, M. Smyth, Heat transfer enhancement in medium temperature thermal energy storage system using a multitube Heat transfer array, Renew. Energy 35 (2010) 198–207.
[80] R. Hendra, T.M.I. Mahlia, H.H. Masjuki, Thermal and melting heat transfer characteristics in a latent heat storage system using mikro, Appl. Therm. Eng. 25 (2005) 1503–1515.
[81] R. Velraj, R.V. Seeniraj, B. Hafner, C. Faber, K. Schwarzer, Heat transfer enhancement in a latent Heat storage system, Sol. Energy 65 (1999) 171–180.
[82] L. Fan, J.M.M. Khodadadi, Thermal conductivity enhancement of phase change materials for Thermal energy storage: a review, Renew. Sustain. Energy Rev. 15 (2011) 24–46. http://dx.doi.org/10.1016/j.rser.2010.08.007.
[83] S.K. Das, S.U. Choi, W. Yu, T. Pradeep, Nanofluids: Science and Technology, John Wiley & Sons, 2007.
[84] J.C. Maxwell, A Treatise on Electricity and Magnetism, Clarendon press, 1881.
[85] W. Yu, D.M. France, J.L. Routbort, S.U.S. Choi, Review and comparison of nanofluid thermal conductivity and heat transfer enhancements, Heat. Transf. Eng. 29 (2008) 432–460.
[86] J. Buongiorno, D.C. Venerus, N. Prabhat, T. McKrell, J. Townsend, R. Christianson, et al., A benchmark study on the thermal conductivity of nanofluids, J. Appl. Phys. 106 (2009) 94312.
[87] C.-W. Nan, R. Birringer, D.R. Clarke, H. Gleiter, Effective thermal conductivity of particulate composites with interfacial thermal resistance, J. Appl. Phys. 81 (1997) 6692–6699.